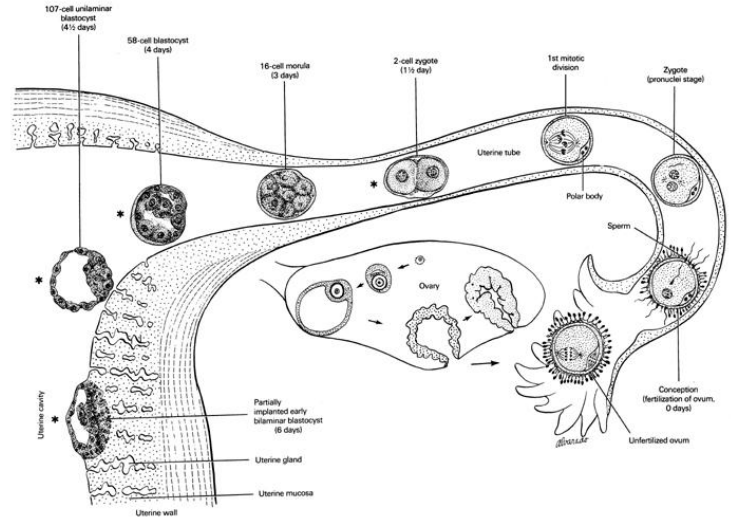
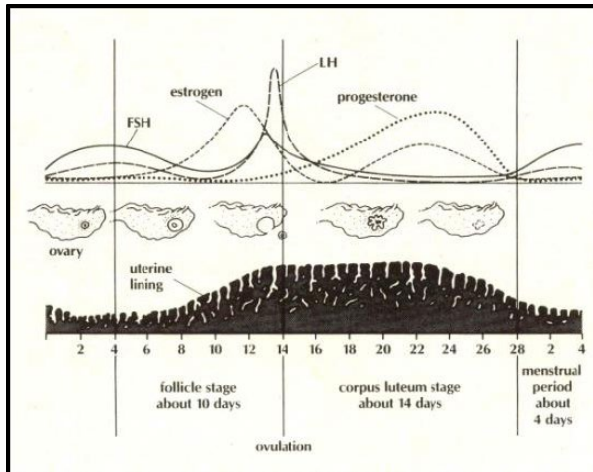


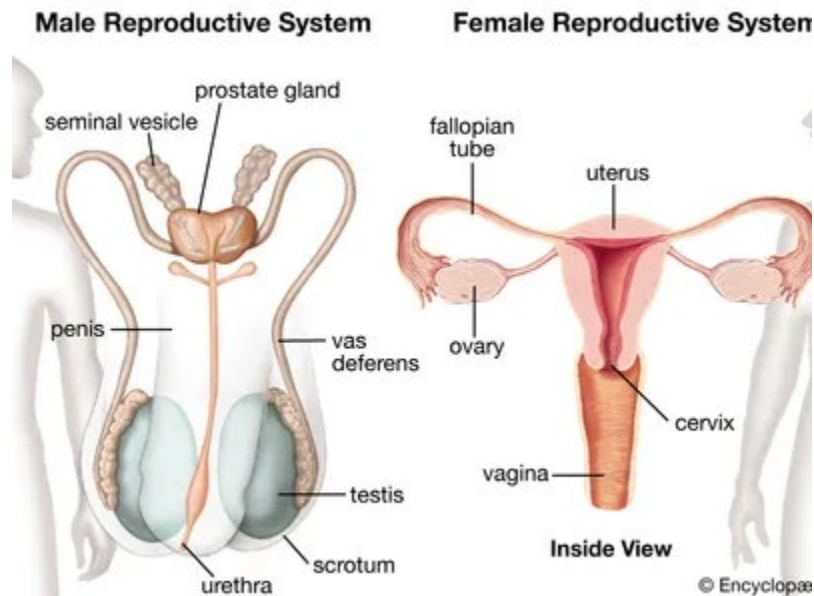


Biology 30

Reproduction and Development Topics 1 and 2 Review Booklet

Name: _____

Topic 5:



Human

Reproductive Systems

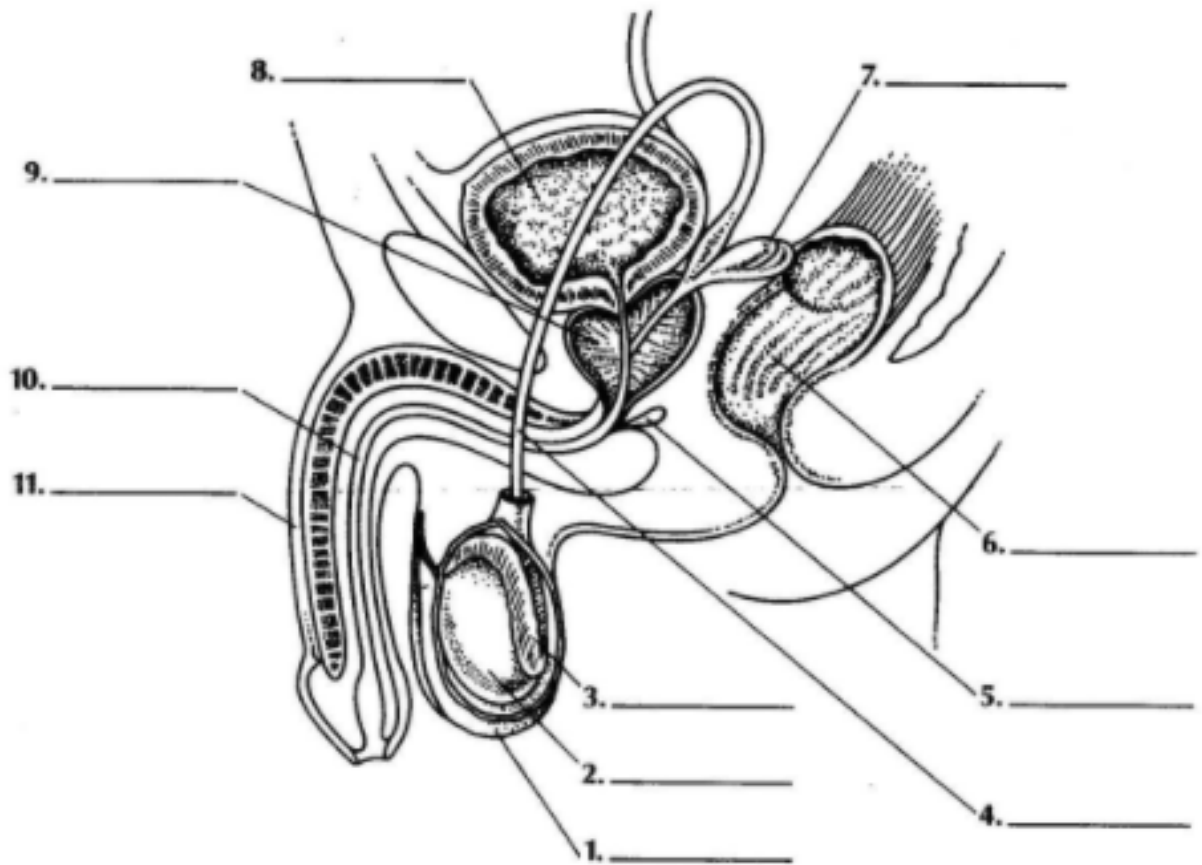
Both the male and female reproductive systems include a pair of gonads:

1. Testes - produce sperm
2. Ovaries - produce eggs

The male and female gonads are responsible for producing sex hormones

1. Testes - produce testosterone and inhibin
2. Ovaries - produce estrogen and progesterone

Label the Male Reproductive System:



Male Reproductive System Structures:

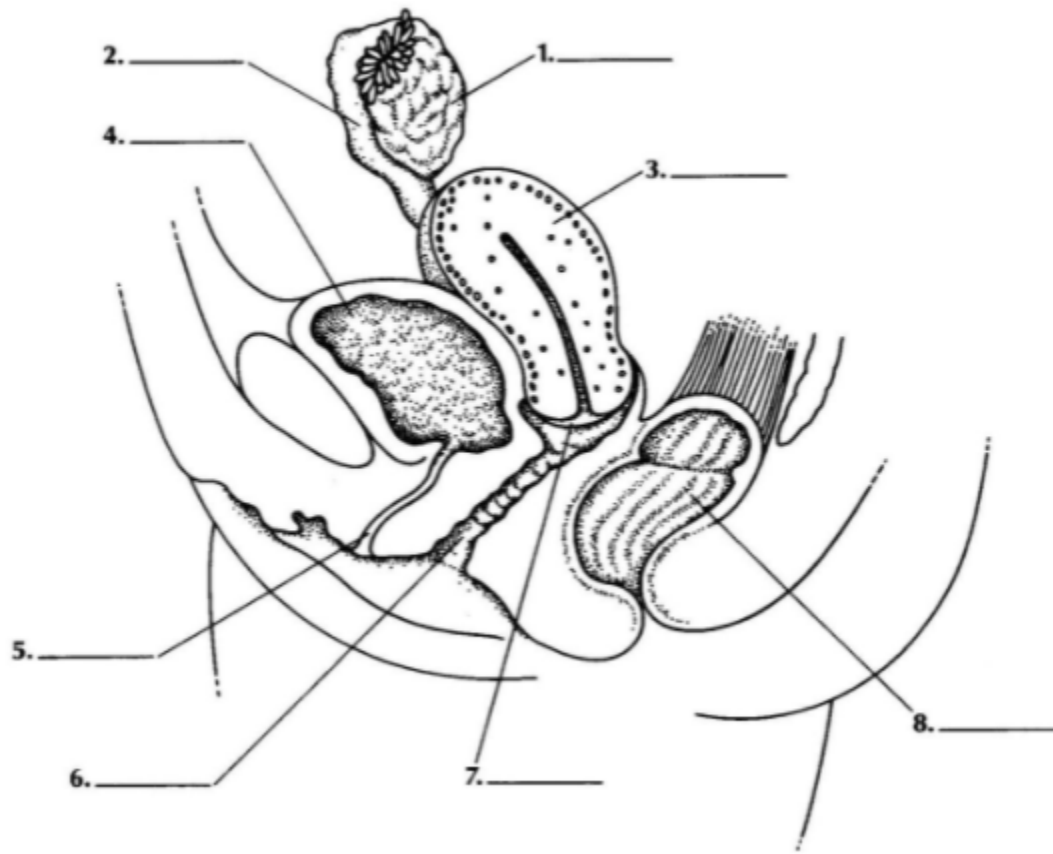
Testes	
Scrotum	
Seminiferous tubules	
Interstitial cells	

Epididymis	
Seminal vesicles	
Prostate gland	
Cowper's gland	

As sperm travels through the vas deferens, they're mixed with fluids from the seminal vesicles, prostate gland and Cowper's gland. The combination of sperm cells and this fluid is called semen.

Describe the pathway of sperm (*where it is made until it leaves the body*)

Label the Female Reproductive System:



Female Reproductive System Structures:

Ovaries	
Follicles	
Fimbriae	
Oviduct (Fallopian tube)	
Uterus	

Endometrium	
Cervix	
Vagina	

Describe the pathway of the egg *(where it is made until it leaves the body)*

What are Sexually Transmitted Infections?

What causes STIs?

What are three ways to prevent STIs?

1. _____
2. _____
3. _____

Compare and contrast bacterial STIs and viral STIs

	Bacterial STIs	Viral STIs
Examples		
Treatment		

Complications for reproductive systems if untreated		

Topic 6: Regulation of the Human Reproductive System

Important terms:

Glands	
Hormones	
Feedback Mechanisms	

Hormonal Regulation of the Male Reproductive System

- Hormonal feedback mechanisms control the process of spermatogenesis and they maintain secondary sexual characteristics of men
- This hormonal feedback begins at puberty and declines at andropause

Hormonal control of the male reproductive system involves five hormones:

Hormone	Gland that Produces it	What is its effect on the body?
Gonadotropin Releasing Hormone (GnRH)		
Follicle-Stimulating Hormone (FSH)		

Luteinizing Hormone (LH)		
Testosterone		
Inhibin		

Male Hormone Feedback Loop - How it works!

1. Hypothalamus releases GnRH
2. Pituitary releases FSH and LH
3. FSH stimulates sperm and inhibin production, LH stimulates testosterone production
4. Increased production of inhibin and testosterone acts on the hypothalamus and pituitary to decrease production of GnRH, FSH, and LH

Hormonal Regulation of the Female Reproductive System

- Hormonal feedback mechanisms control the process of oogenesis and they maintain secondary sexual characteristics of men
- This hormonal feedback begins at puberty and ends at menopause

Hormonal control of the female reproductive system involves five hormones:

Hormone	Gland that Produces it	What is its effect on the body?
Gonadotropin Releasing Hormone (GnRH)		

Follicle-Stimulating Hormone (FSH)		
Luteinizing Hormone (LH)		
Estrogen		
Progesterone		

Female Hormone Feedback Loop - How it works!

- Hormonal regulation of the female reproductive system is called the menstrual cycle
- The menstrual cycle is actually two separate but interconnected cycles of events:

1. **Ovarian Cycle**
2. **Uterine Cycle**

1. Menstrual Cycle - The Ovarian Cycle

Divided into two stages:

Follicular Stage

--

Luteal Stage

2. Menstrual Cycle - The Uterine Cycle

--

If fertilization does NOT occur the uterine lining breaks down, blood and tissue pass out of the uterus through the vagina. This is called the woman's PERIOD or MENSTRUATION

For each of the following explain what happens during the Menstrual Cycle.

a) hormone levels (include FSH, LH, Estrogen and Progesterone)

b) the endometrial lining

c) the follicle developing in the ovary
